

PCA

Why do we need pca.

* Curse of dimensionality: with increase in no. of features performance of model degrades

dimension $\Rightarrow$ features

M_1	M_2	M_3	M_4	M_5	M_6	M_7
20f	50f	100f	250f	500f	750f	1000f
$Acc/R^2 \uparrow$	$Acc/R^2 \uparrow$	$Acc/R^2 \uparrow$	$Acc/R^2 \uparrow$	$Acc/R^2 \approx$	$Acc/R^2 \approx$	$Acc/R^2 \approx$

with increase in no. of features, after one point of time the accuracy / R^2 will not increase in that proportion

Why? $\rightarrow$ • few of feature has multicollinear.

Analogy $\rightarrow$ if 20 features giving 30% acc and 100 giving 33%. So it is not worth that much computation cost.

- few of features might be exactly same
- no. variance/info in feature.
- lot of duplicate entries.
- No business value.

To remove Curse of dimensionality $\leftarrow$ ① feature selection
② feature extraction $\rightarrow$ PCA

* Let's say out of 1000 we took 50 features whose correlation is highest, but we are losing a lot of data. So when we want info of 1000 feature at the same time we want to remove curse of dimensionality. we have this feature extraction technique (PCA).

- * PCA Geometrical intuition
It is about learning patterns from the data.

Since PCA is dimensionality reduction technique $\rightarrow$ we will try to reduce the dimension.

if you want to convert $2\Omega \rightarrow 1\Omega$

↳ but here we are losing 2 features, which is happen in feature selection.

- To get both the features we need to understand the concept of variance spread/ information

spread $\rightarrow$ when we regulate $PQ_2 \rightarrow$ less amount of variance is lost.

A graph illustrating the relationship between PC1 and PC2. The y-axis is labeled PC2 and the x-axis is labeled PC1. A red line connects two points, labeled 'new PC1' and 'vari'. A blue line points to the y-axis, labeled 'this is lost'. A dashed blue line points to the x-axis, labeled 'vari'.

$x_1, x_2, x_3, \dots, x_{1000}$	y	PC_1	PC_2	$PC_3 \dots PC_{1000}$	Principal Component will be equal to no. of features
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Note $\neq$ $Varianc(PC_1) > Var(PC_2) \dots > Var(PC_3)$

first few PC's will capture almost 90% variance of data. So instead of 1000 PC's you will use first few

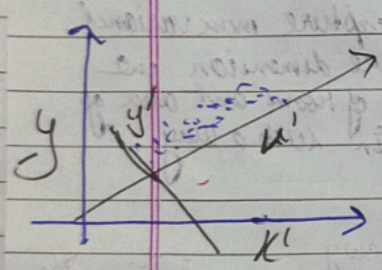
Principal Component y (target variable) is not used, that's why PCA is also called as unsupervised algorithm.

How axis transformation happens?

→ This is done by Eigen decomposition of covariance matrix

Disadvantage of PCA → loss of information

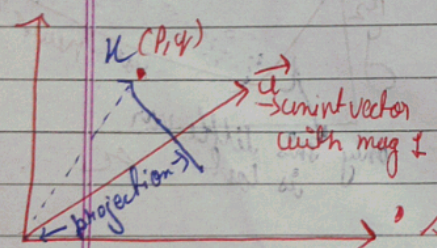
MATHEMATICAL EXPLANATION OF P.C.A



- ① How to get k', y'
- ② How to get ~~for~~ projection on k'

Projection of k on $u = \frac{k \cdot u}{||u|| \cdot ||k||}$

→ $k \cdot u$ → scalar value or projection



→ Similarly for n datapoints we have n projections $d_1, d_2, d_3 \dots d_n$

NOTE

• Now we have $(d_1, d_2, d_3, \dots, d_n) \rightarrow$ and we want that unit vector where variance/spread is maximum.

$d_1, d_2, d_3, \dots, d_n \rightarrow$ are projection of diff. dp's

• we want maximum variance.

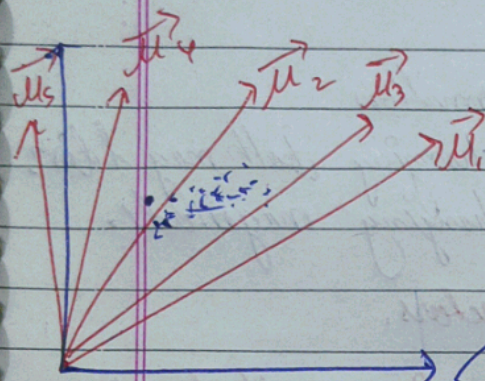
$$\text{Var} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

↓
spread of dp with itself

aim is to find that unit vector with highest variance, so here var is used as cost fun.

• we have multiple features so variance becomes covarianced

$$\text{Covar} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n}$$



• Now with help of co-variance we will create a co-variance matrix as we have n-no. of features.

• it helps us to get that unit vector which have maximum spread.

Co-variance matrix $\rightarrow$ $\begin{matrix} & u_1 & u_2 \\ \text{Variance - Co-variance matrix} & \begin{matrix} u_1 & u_2 \end{matrix} \end{matrix} \begin{matrix} \text{Cov}(u_1, u_1) & \text{Cov}(u_1, u_2) \\ \text{Cov}(u_2, u_1) & \text{Cov}(u_2, u_2) \end{matrix}$

$\begin{matrix} & u_1 & u_2 \\ u_1 & \text{Var}(u_1) & \text{Cov}(u_1, u_2) \\ u_2 & \text{Cov}(u_2, u_1) & \text{Var}(u_2) \end{matrix} \rightarrow \text{Similarly } \begin{matrix} & u_1 & u_2 & u_3 \\ u_1 & \text{Var}(u_1) & \text{Cov}(u_1, u_2) & \text{Cov}(u_1, u_3) \\ u_2 & \text{Cov}(u_2, u_1) & \text{Var}(u_2) & \text{Cov}(u_2, u_3) \\ u_3 & \text{Cov}(u_3, u_1) & \text{Cov}(u_3, u_2) & \text{Var}(u_3) \end{matrix}$

NOTE $\rightarrow$ Theorem $\rightarrow$ • If you decompose a covariance matrix of features, then you will get eigen value and eigen vector.
• Eigen Vector of highest magnitude of eigen value captures maximum variance/spread.

But what is decomposition?

→ decomposition is some mathematical transformation

Linear transformation of a matrix $\Rightarrow$ it changes both

$$A \cdot \vec{V} = \lambda \cdot \vec{V}$$

magnitude and direction of a vector.

$A \Rightarrow$ matrix

$V \rightarrow$ eigenvector

$\lambda \rightarrow$ eigen value $\rightarrow$ scalar value

A matrix transformation which brings changes in coordinate system.

$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \rightarrow$ matrix $\rightarrow$ linear transformation

$\rightarrow$ many vectors are changing both mag & dir.

$\rightarrow$ some vectors only changing magnitude.

These vectors are eigen vectors.

eigen values are change in magnitude for those eigen vectors

* The vectors changes only magnitude will be equal to dimension of matrix / no. of features

$\hookrightarrow$ These vectors are also called as principal component

$\hookrightarrow$ And ~~max~~ variance captured is that eigen value for eigen vector.